

Test Case ID	Test Description	Input Data	Expected Output	Actual Output	Status
TC_01	User Registration	Valid user details	User registered successfully	User registered successfully	1
TC_02	User Registration	Invalid email format	Error message: "Invalid email format"	Error message: "Invalid email format"	1
TC_03	Anemia Prediction	Valid CBC parameters	Anemia prediction result	Anemia prediction result	1
TC_04	Anemia Prediction	Missing CBC parameters	Error message: "Please fill out this field."	Error message: "Please fill out this field."	1
TC_05	Login Authentication	Valid username and password	Redirect to dashboard	Redirect to dashboard	
TC_06	Login Authentication	Invalid username or password	Error message: "Invalid credentials"	Error message: "Invalid credentials"	1

9. Future Enhancements

Future improvements could include:

- Incorporating additional machine learning models for comparison.
- Enhancing the dataset with more diverse and comprehensive data.
- Adding features for real-time data input from electronic health records.
- Developing a mobile application for easier access by patients and healthcare providers.

Overall Precision, Accuracy, and Error

Overall Accuracy

Overall accuracy is the ratio of correctly predicted instances (both true positives and true negatives) to the total number of instances.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

In a confusion matrix, overall accuracy is calculated as:

$$\text{Accuracy} = \frac{\sum \text{Diagonal Elements}}{\sum \text{All Elements}}$$

For our confusion matrix:

$$\text{Accuracy} = \frac{80+34+5+3+1+56+45+13+17}{80+1+34+5+3+1+56+45+1+13+1+17}$$

Calculating the sums:

$$\text{Numerator (TPs)} = 254$$

$$\text{Denominator (Total Instances)} = 258$$

$$\text{Accuracy} = \frac{254}{258} \approx 0.984 \text{ (or 98.4\%)}$$

Overall Precision

Overall precision can be calculated as the weighted average of the precision of each class, weighted by the number of instances of each class.

$$\text{Precision} = \frac{\sum (\text{Precision for each class} \times \text{Number of instances for that class})}{\sum \text{Number of instances for each class}}$$

Calculating class-wise precision again:

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- Class 0: 1.00
- Class 1: 0.971
- Class 2: 0.833
- Class 3: 1.00
- Class 4: 1.00
- Class 5: 1.00
- Class 6: 0.978
- Class 7: 0.929
- Class 8: 1.00

Number of instances for each class:

- Class 0: 81
- Class 1: 34
- Class 2: 6
- Class 3: 3
- Class 4: 1
- Class 5: 56
- Class 6: 46
- Class 7: 14
- Class 8: 17

Using these values to calculate weighted precision:

Using these values to calculate weighted precision:

$$\text{Overall Precision} = \frac{(1 \times 81) + (0.971 \times 34) + (0.833 \times 6) + (1 \times 3) + (1 \times 1) + (1 \times 56) + (0.978 \times 46) + (0.929 \times 14) + (1 \times 17)}{81 + 34 + 6 + 3 + 1 + 56 + 46 + 14 + 17}$$

$$\text{Overall Precision} = \frac{81 + 33.014 + 4.998 + 3 + 1 + 56 + 44.988 + 13.006 + 17}{258}$$

$$\text{Overall Precision} = \frac{253.006}{258} \approx 0.980 \text{ (or 98.0\%)}$$

Overall Error

Overall error is the complement of accuracy:

$$\text{Error} = 1 - \text{Accuracy}$$

$$\text{Error} = 1 - 0.984$$

$$\text{Error} = 0.016 \text{ (or 1.6\%)}$$

Summary

- Overall Accuracy: 98.4%
- Overall Precision: 98.0%
- Overall Error: 1.6%

These metrics indicate that the model performs very well in predicting the types of anemia, with high accuracy and precision, and a low error rate.

Confusion Matrix

